

JEE MAIN REVISION MASTER: MOVING CHARGES AND MAGNETISM

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 48 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Biot-Savart Law & Ampere's Circuital Law	28 PYQs	Magnetic Field
Very High	Lorentz Force, Cyclotron & Magnetic Force on Wires	21 PYQs	Magnetic Forces

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Biot-Savart Law & Ampere's Circuital Law	$dB = \frac{\mu_0 I dl \sin\theta}{4\pi r^2}, \quad B_{\text{wire}} = \frac{\mu_0 I}{2\pi d}, \quad B_{\text{center}} = \frac{\mu_0 N I}{2R}, \quad B_{\text{solenoid}} = \mu_0 n I$
Lorentz Force, Cyclotron & Magnetic Force on Wires	$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}), \quad r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}, \quad T = \frac{2\pi m}{qB}, \quad \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Biot-Savart Law & Ampere's Circuital Law (Very High Yield — 28 PYQs)

Core Principle: Calculation of magnetic fields generated by steady currents in wires, loops, and solenoids.

■■ **Caution / JEE Pitfall:** In solenoid formula $B = \mu_0 n I$, n is number of turns per UNIT LENGTH ($n = N/L$), not total turns N .

■ **Speed Trick:** Right-hand thumb rule: thumb points along current, curled fingers give direction of circular magnetic field lines.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this cha..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Two identical circular current loops carrying equal currents are placed with their axes inclined at 45° to each other as shown in t..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Biot-Savart law indicates that an electron moving with a velocity $\vec{v}$ produces a magnetic field $\vec{B}$ around it such that..." [Ans: N/A]

3.2 Lorentz Force, Cyclotron & Magnetic Force on Wires (Very High Yield — 21 PYQs)

Core Principle: Trajectory of charged particles in magnetic fields, velocity selector ($\vec{E} \perp \vec{B}$), and force between parallel currents.

■ **Caution / JEE Pitfall:** Magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$ does ZERO work because $\vec{F} \perp \vec{v}$. It changes direction of velocity, never kinetic energy!

■ **Speed Trick:** Parallel currents in the SAME direction ATTRACT; anti-parallel currents REPEL.

Representative Source PYQs:

- **[2026 Shift] Q#4 (MCQ):** "A proton, an electron and an α -particle enter at right angles to a uniform magnetic field with the same velocity. If R_p, R_e a..." **[Ans: N/A]**
- **[2025 Shift] Q#1 (MCQ):** "A square loop of side 2 m lies in the Y-Z plane in a region having a magnetic field $\vec{B} = (5\hat{i} + 3\hat{j} - 4\hat{k})T$. The magnitude ..." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "19. A particle of specific charge $q/m = \pi \text{ Ckg}^{-1}$ is projected the origin towards positive X-axis with the velocity 10ms^{-1} in a u..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this charge. Then

- (A) observes both electric and magnetic fields
- (B) observes both electric and magnetic fields
- (C) observes only magnetic field
- (D) observes only electric field

Correct Answer: None

Solution Analysis: The correct option is: Option B: B observes both electric and magnetic fields The point charge is placed in the moving train. For passenger A sitting inside the train, the charge is at rest. A stationary charge produces only an electric field around it. So A observes only electric field, not magnetic field. For person B standing on the ground, the train is moving. So the charge is also moving with the train. A moving charge is equivalent to an electric current, and current produces a magnetic field. Since the charge itself also has electric field, B observes both electric and magnetic

Benchmark Problem 2: KCET 2026 (Q#4)

A proton, an electron and an α -particle enter at right angles to a uniform magnetic field with the same velocity. If R_p, R_e and R_α are the radii of circular paths of these particles, then

- (A) $R_\alpha = R_p = R_e$
- (B) $R_\alpha > R_p > R_e$
- (C) $R_\alpha < R_p < R_e$
- (D) $R_\alpha > R_p = R_e$

Correct Answer: None

Solution Analysis: For a charged particle moving at right angles to a uniform magnetic field, the magnetic force provides the centripetal force. So, $qvB = \frac{mv^2}{R}$ This gives $R = \frac{mv}{qB}$ Here, all three particles enter with the same velocity v in the same magnetic field B . So the radius depends on (m/q) For proton: $R_p = \frac{m_p v}{qB}$ For electron: $R_e = \frac{m_e v}{qB}$ Since the electron has the same magnitude of charge as the proton but much smaller mass, $R_e < R_p$ For α -particle: An α -particle has mass nearly $4m_p$ and charge $2e$. $R_\alpha = \frac{4m_p v}{2qB} = \frac{2m_p v}{qB} = 2R_p$